

PHI 347 Neuroscience and Philosophy (3 cr)

TuTh 4:30-5:45 pm Location: Withers 120

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Office Hours: W 1:30-2:30 PM and by appointment, Withers Hall 453

Catalog Description: Introduction to principal theoretical, empirical and normative issues at the intersection of neuroscience and philosophy, including such issues as: consciousness, the mind's "I" and the brain's "I"; free will, moral responsibility and neuroscience; the ethics of personal enhancement; brains, human nature and personal identity; neuroscientifically informed evaluation of well-being.

Neuroscience is making significant discoveries and developments at a rapid pace and increasing our knowledge of the mind/brain. This course is intended to introduce students to the fields of inquiry at the intersection between neuroscience and philosophy, including its normative, descriptive and theoretical foci and methods. The course is designed so that students will gain both knowledge and skills in practical and theoretical issues. Students will navigate general review texts, as well as empirical and normative texts during the course. Some of these will be oriented towards describing the science, while others lean towards conceptual analysis and normative claims arising from the use of neuroscientific knowledge. Furthermore, the learning experiences should be enhanced through discussions, critical evaluation of conceptual and empirical papers, and an essay exam at the end.

Pre-requisites, co-requisites, enrollment restrictions: none

GEP: satisfies 3 cr hrs of either Humanities or Interdisciplinary Perspectives requirements

GEP Humanities Objectives and Outcomes

This course will provide instruction and guidance that help students to:

1. Engage the human experience through the interpretation of human culture and
2. Become aware of the act of interpretation itself as a critical form of knowing in the humanities; and
3. Make academic arguments about the human experience using reasons and evidence for supporting those reasons that are appropriate to the humanities.

Outcomes:

For Objective 1: Students will be able to describe a variety of philosophical conceptions of the mind, self and consciousness addressed by contemporary cognitive neuroscience and to assess the plausibility of different approaches taken by the sciences in attempting to explain those philosophical conceptions.

Measures for 1: Essays on lecture topics, Midterm and Final exam questions.

(Sample question: How has mind/body dualism informed judgments of moral responsibility? What would be the effect on those judgments of rejecting dualism in favor of an alternative view grounded in cognitive neuroscience?)

For Objective 2: Students will be able to apply philosophical methods to evaluate the application of neuroscientific explanations to ethical and social implications (e.g. Are arguments intended to uncover truth or overcome opponents?).

Measures for 2: Essays on lecture topics, Midterm and Final exam questions.

(Sample question: How do some of what's been learned about the human central nervous system promise or threaten to challenge traditional valuing of altruistic behavior and motivating compassion?)

For Objective 3: Analyze and reconstruct neuroscientific arguments concerning the supposed basis of the mind in the brain and the alleged further implications for ethics.

Measures for 3: Essays on lecture topics, Midterm and Final exam questions.

(Sample question: Are there some kinds of neurological disorders that would under some circumstances absolve a person suffering from them of moral responsibility for particular kinds of actions? What would be a good example of neurological disorders, circumstances and actions in question?)

GEP Interdisciplinary Perspectives Objectives and Outcomes:

The course will also provide instruction and guidance in interdisciplinary perspectives that help students to:

1. Distinguish between the distinct approaches of philosophy and neuroscience; and
2. Identify and apply authentic connections between philosophy and neuroscience; and
3. Explore and synthesize the approaches or views of philosophy and neuroscience.

For Objective 1: Students will be able to identify and differentiate the contributions of neuroscience and philosophy to the development of important questions concerning meaning and responsibility.

Measures for 1: Performance on in-class presentation, essays on lecture topics, mid-term exam and the final exam. (Sample question: What are the ethical implications of claims from neuroscience about the neural basis of conscious volition or free will?)

For Objective 2: Students will be able to describe and analyze the strengths and weaknesses of the neurophilosophical model of the mind/brain both for philosophical rigor and empirical grounding in neuroscience.

Measures for 2: Performance on in-class presentation, essays on lecture topics, mid-term exam and the final exam. (Sample question: How does the integrative neurophilosophical model of the mind/brain explain or explain away traditional intuitions about the nature of consciousness and subjective experience?)

For Objective 3: Students will be able to distinguish, analyze and critically evaluate theoretical, normative and empirical claims in neuroscience and philosophy.

Measures for 3: Performance on the mid-term exam, final exam topical focus proposal and the final exam itself. [Sample question: Identify three methodologically and ethically debatable uses of neuroimaging and say how they relate to the historical development of brain science (e.g., phrenology).]

[Disciplines synthesized, connected and considered: Neuroscience, Philosophy;

Method of integrating multiple points of view into a cohesive understanding: The instructor will guide students in examining old questions about the nature of the mind within the new framework of the brain sciences. The neurobiological basis of consciousness, the self, and

free choice will be addressed as well as the current state of knowledge about how the brain learns about the external world and about its own introspective world. These questions necessarily entail an attempt to develop an integrated approach. Drawing on results from research at the neuronal, neurochemical, system, and whole-brain levels the course gives an up-to-date perspective on the basis and significance of religious and moral experiences. The course instructor encourages critical thinking at all stages of reconstructing the current state of neurophilosophy - analyzing what we know, describing what we do not know, and identifying where things may go from here.]

Course Student Learning Outcomes:

By the end of the course, students will be able to:

1. Identify major empirical, normative (ethical and non-ethical) and methodological issues in the intersection of neuroscience and philosophy;
2. Determine which approaches are effective in analyzing these issues;
3. Differentiate among levels of explanation and modeling in neuroscience as the levels affect potential practical applications of any neuroscientific hypotheses;
4. Distinguish empirical from theoretical justifications (both of which are typically needed) in evaluating proposed resolutions of these issues.

Texts:

Patricia Smith Churchland, *Brain-Wise: Studies in Neurophilosophy*, (The MIT Press, 2002). ISBN 978-0262532006 paperback \$ 36.46 (new) \$ 3.99 (used) and available for download at: [https://archive.org/details/Brain-Wise Studies in Neurophilosophy by Patricia Smith Churchland](https://archive.org/details/Brain-Wise_Studies_in_Neurophilosophy_by_Patricia_Smith_Churchland) [Required]

Additional required readings will be available online through the course Moodle web site (<https://wolfware.ncsu.edu>). See Schedule below for details.

Fall 2024 Schedule (unless noted otherwise, readings are in *Brain-Wise*) Readings should be completed by the date indicated.

Date			Topics & Required Readings	Tests & Assignments
Aug	20	Tu	Course overview and introduction; instructor assigns in-class presentation topics and dates	
Aug	22	Th	Core questions (pp. 1-5)	
Aug	27	Tu	Natural philosophy (pp. 5-20)	In-class presentations begin; essays on lecture topics may be submitted
Aug	29	Th	Library Workshop w. Cynthia Levine	
Sep	3	Tu	Reductions and Coevolution in Scientific Domain (pp. 20-30) AND Hypotheses of "Brain-Wise" (pp. 30-34)	
Sep	5	Th	NeuroComputational Ethics Research in V.R.	
Sep	10	Tu	Intro to Metaphysics (pp. 37-43) AND Metaphysics and the Mind (pp. 43-50)	

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Date			Topics & Required Readings	Tests & Assignments
Sep	12	Th	Self & Self-Knowledge 1: The Problem (pp. 59-90)	
Sep	17	Tu	Wellness day (no classes, no readings)	
Sep	19	Th	Self & Self-Knowledge 2: Inner Models of Body, Self, and Others (pp. 91-125)	
Sep	24	Tu	Consciousness 1: The Problem and Empirical Directions (pp. 127-170)	
Sep	26	Th	Consciousness 2: Dualism and Arguments against Neuroscientific Progress (pp. 171-199)	
Oct	1	Tu	Normative implications of Neuroscience (Joshua Greene, "From neural 'is' to moral 'ought': What are the moral implications of neuroscientific moral psychology?", <i>Nature Reviews Neuroscience</i> , 4 (2003), 847-851; Greene, J. D., Sommerville, R. B., Nystrom, L. E., Darley, J. M. & Cohen, J. D. (2001): An fMRI investigation of emotional engagement in moral judgment, <i>Science</i> 293: 2105 – 2108.)	
Oct	3	Th	Cognitive Neuroscience and the Dual systems model of moral judgment (Greene, J. D. (2008): "The secret joke of Kant's soul," in Sinnott-Armstrong, W. (Ed.): <i>Moral psychology Vol. 3, The neuroscience of morality</i> , Cambridge, MA: MIT Press; 35-79.)	
Oct	8	Tu	Philosophical reactions: The normative insignificance of neuroscience (Berker, S. (2009): "The normative insignificance of neuroscience," <i>Philosophy and Public Affairs</i> , 37 (4): 293-329.)	
Oct	10	Th	Free Will, Responsibility, and Neuroscience (pp. 201-236)	
Oct	15	Tu	Fall break (no classes, no readings)	
Oct	17	Th		In-class Midterm exam
Oct	22	Tu	Interpretation of Neuroscientific Findings on Volition: Libet & Haggard (Haggard, P. and Libet, B. (2001): Conscious intention and brain activity, <i>Journal of Consciousness Studies</i> , 8 (11): 47-63.)	
Oct	24	Th	The integrative approach to neuroscience of ethics (Dubljević, V. & Racine, E. (2014): "The ADC of Moral Judgment: Opening the Black Box of Moral Intuitions with Heuristics about Agents, Deeds and Consequences," <i>AJOB – Neuroscience</i> , 5 (4): 3-20;	

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			Dubljević, V., Sattler, S., & Racine E. (2018): Deciphering moral intuition: How agents, deeds and consequences influence moral judgment, <i>PLOS One</i> , doi:10.1371/journal.pone.0204631.)	
Oct	29	Tu	A Post-metaphysical position on autonomy, freedom and responsibility (Dubljević, V. (2013): Autonomy in Neuroethics: Political and not Metaphysical; <i>American Journal of Bioethics (AJOB) – Neuroscience</i> , 4 (4): 44-51.)	
Oct	31	Th	Neuroscience and Epistemology (pp. 241-272)	Take Home Final Exam proposal and references due by beginning of class
Nov	5	Tu	Representation in the Brain (pp. 273-318)	
Nov	7	Th	Brain Learning (pp. 321-369)	
Nov	12	Tu	Guest Lecture by Dr. Shaun Respass (TBA)	
Nov	14	Th	Religion and the Brain (pp. 373-402)	
Nov	19	Tu	Brain-based ethics? (Roskies, A. (2002): Neuroethics for the New Millenium, <i>Neuron</i> , 35: 21-3. ; Levy, N. (2011): Neuroethics, a new way of doing ethics, <i>AJOB-Neuroscience</i>)	
Nov	21	Th	Other perspectives: Critical Neuroscience (Vidal, F. (2015): Historical and ethical perspectives of modern neuroimaging, in Clausen, J. & Levy, N. (Eds.): <i>Handbook of Neuroethics</i> , Dodrecht: Springer, pp. 535-350.)	
Nov	26	Tu	Neuroscience and Personhood (Martha J Farah and Andrea S Heberlein, "Personhood and Neuroscience: Naturalizing or Nihilating?," <i>AJOB-Neuroscience</i> , 7 (2007), 37-48.)	
Nov	28	Th	Thanksgiving day (no classes, no readings)	
Dec	3	Tu	Frontiers of Neuroscience: Addiction (Steven E Hyman, "The Neurobiology of Addiction: Implications for Voluntary Control of Behavior.," <i>AJOB-Neuroscience</i> , 7 (2007), 8-11.)	
Dec	4	W	Reading day	
Dec	5	Th	Take Home Final Exam (due by 6:00 PM at WI 453)	

The course schedule is subject to change with appropriate notification to students.

Grading:

Required work	Points
1) Midterm Exam (25%) - in-class, five short essay questions	25
2) Participation (10%) (see below)	10
3) In-class presentation (10-15 slides or equivalent, which need to be submitted via e-mail 24 hours before the assigned class period) of key issues in the assigned reading material, with topic and date as assigned on Aug 20 (10%)	10
4)* 5-page essay (with at least 5 external references which are not included in the page count) on the discussion topic from one assigned lecture after first two classes - due by the end of class one week after the chosen lecture, submitted BOTH via e-mail and a hard copy by 4:30 PM (20%)	20
5)* Proposal for topical focus of cooperatively designed Final Exam: Descriptive title, 250 words proposal, plus list of properly cited suggested literature - at least 10 high quality peer-reviewed articles, book chapters or equivalent (10%) due by Oct 31, by the beginning of class.	10
6)* Final Exam (25%): Take home, essay format, exactly 10 pages, 12 point font, 1" margins (excluding title page & references) due Dec 5, 6:00 PM at WI 453.	25
*For items 4), 5) and 6) BOTH electronic and <i>hard copies</i> must be submitted with deadlines as indicated in the schedule above.	

Participation: One of the following assessments will be made of students' preparation and participation on a weekly basis. The weekly assessments will be averaged to determine the final participation grade. Any day that you do not attend class and for which your absence is not excused (see <https://policies.ncsu.edu/regulation/reg-02-20-03>) or on which you are 15 minutes or more late for class, you will be recorded as absent and will receive no points for participation on that day.

A+ (100%) Student's participation is outstanding in both quality and quantity. Student demonstrates a full mastery of the course material.

A (95%) Student is well prepared, attentive, always responds when called upon and volunteers often with pertinent answers or questions. Student demonstrates a strong facility with the course material and complex structures, but makes minor mistakes.

B (85%) Student is usually prepared, responds when called on and volunteers on occasion. Student demonstrates a basic familiarity with assigned material, but falters with complex structures.

C (75%) Student shows evidence of being unprepared on occasion, has trouble when called on and does not volunteer often. Student demonstrates some familiarity with assigned material, but occasionally falters on basic structures.

D (65%) Student is unprepared, inattentive, never volunteers, or comes to class late. Student demonstrates occasional familiarity with assigned material, but regularly falters on basic structures.

F (0%) Student exhibits a lack of concern for the class, sleeps in class, or disturbs the class. Student fails to show familiarity with assigned material.

The course letter grade will be determined according to standard university letter grading:

$97 \leq A+ \leq 100$	$77 \leq C+ < 80$
$93 \leq A < 97$	$73 \leq C < 77$

$90 \leq A- < 93$	$70 \leq C- < 73$
$87 \leq B+ < 90$	$67 \leq D+ < 70$
$83 \leq B < 87$	$63 \leq D < 67$
$80 \leq B- < 83$	$60 \leq D- < 63$
	$0 \leq F < 60$

Scores with a fractional part ≥ 0.50 points will be rounded up.

Attendance/Absence Policy: Attendance is required and will be recorded each class period. You may miss two classes without a documented excuse. After that threshold, your final grade will be penalized. Specifically, unexcused absences are calculated as follows: 3 absences (final grade lowered by a third, e.g. from an A to an A-); 4 absences (final grade lowered by two thirds, e.g. from an A to an B+); 5 absences (final grade lowered by a whole letter grade, e.g. from an A to B); 6 or more absences (*automatic F for the course*, regardless of work completed). Excused absences, following the university definition of an excused absence, must be documented properly. Please review the university attendance regulation at <http://policies.ncsu.edu/regulation/reg-02-20-03> for further information, including the university definition of excused absences. Please contact the instructor immediately if you have a conflict to arrange a deadline for a makeup assignment (e.g., a 750-word summary of required readings for each missed class, regardless of reason).

Late Assignments. Late work will be marked down one full letter grade per day. Extensions will be granted only in the case of excused absences, in accord with the university attendance policy cited above.

Academic Integrity

In all course work, students are bound by the Pack Pledge. [*"I have neither given nor received unauthorized aid on this test or assignment"*]. Any violation of this Pledge will result in referral to the Office of Student Conduct with a recommendation of a zero/F for at least the affected assignment; a failing grade for the course may be imposed as may other penalties. See the **Code of Student Conduct policy (NCSU POL11.35.1)** <https://policies.ncsu.edu/policy/pol-11-35-01>

Disability Accommodations

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office on the third floor of Holmes Building (Suite 304). For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (REG02.20.01) <https://policies.ncsu.edu/regulation/reg-02-20-01>.

Privacy

Students may be required to disclose personally identifiable information to other students in the course, via electronic tools like email or web-postings, where relevant to the course. Examples include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

N.C. State Policies, Regulations, and Rules (PRR)

Students are responsible for reviewing the NC State University PRR's which pertains to their course rights and responsibilities:

- **Equal Opportunity and Non-Discrimination Policy Statement** <https://policies.ncsu.edu/policy/pol-04-25-05> with additional references at <https://oied.ncsu.edu/equity/policies/>
- **Code of Student Conduct** <https://policies.ncsu.edu/policy/pol-11-35-01>
- **Grades and Grade Point Average** <https://policies.ncsu.edu/regulation/reg-02-50-03>
- **Credit-Only Courses** <https://policies.ncsu.edu/regulation/reg-02-20-15>
- **Audits** <https://policies.ncsu.edu/regulation/reg-02-20-04>
- **Incompletes (IN)** <https://policies.ncsu.edu/regulation/reg-02-50-03>

Department web site: <https://philrel.chass.ncsu.edu/>

Supporting Fellow Students in Distress:

As members of the NC State Wolfpack community, we each share a personal responsibility to express concern for one another and to ensure that this classroom and the campus as a whole remains a healthy and safe environment for learning. Occasionally, you may come across a fellow classmate whose personal behavior concerns or worries you, either for the classmate's well-being or yours. When this is the case, I would encourage you to report this behavior to the NC State's Students of Concern website: <http://go.ncsu.edu/NCSUcares>. Although you can report anonymously, it is preferred that you share your contact information so they can follow-up with you personally.

Use of electronic devices in class: Use of laptops, tablets, cellphones, smartwatches or any other communication or AI-powered devices during class or for class assignments is not allowed unless special permission has been obtained. All cellphones, smartwatches and any other devices capable of producing sound must be silenced.

ClassEval Schedule: Online class evaluations will be available for students to complete during the last 2 weeks of the semester then become unavailable at 8 am on the first day of finals.

Students will receive an email message directing them to a website where they can login using their Unity ID and complete evaluations. All evaluations are confidential; instructors will not know how any one student responded to any question, and students will not know the ratings for any instructors.

Evaluation website: <https://go.ncsu.edu/cesurvey> or
<https://oirp.ncsu.edu/surveys/classeval/for-students>

Student help desk: classeval@ncsu.edu

More information about ClassEval: <https://oirp.ncsu.edu/surveys/classeval>

If you are taking this course to meet a GEP requirement *do not take it as Credit Only (S/U)*.

Restriction on Sharing Content of Course Materials

Copying, distributing, downloading, and uploading information on the Internet may infringe the copyright for that information. Even an innocent, unintentional infringement violates the law. Violations of copyright law that occur on or over the university's networks or other computer resources may create liability for the university as well as the computer user. <https://policies.ncsu.edu/regulation/reg-01-25-02>

Accordingly, repeat infringers will have their computer account and other access privileges terminated by NC State. By accessing e-reserve and course web site material you agree to the following statement: "I acknowledge that all documents¹ made available to me for the course PHI 347 taught at NC State University in the current term are copyrighted and intended only for my personal use. By logging into the PHI 347 web site, you certify that you will not share any content of the class PHI 347 NC State University website with a third party² without written permission from the course instructor Professor Veljko Dubljevic."

REG 02.20.11 - Recording Lectures and Discussions Students may not use recording devices in the classroom without explicit prior permission of the instructor. If permission is granted, there must also be no member of the class who objects. Instructor and class permission is not required when an accommodation notification from Disability Services has been received by the instructor, which identifies a student that requires the use of a recording device. However, the instructor may prohibit the use of any recording device when it would inhibit free discussion and free exchange of ideas in the classroom.
<https://policies.ncsu.edu/regulation/reg-02-20-11>

CAREER SERVICES: Career Services <https://www.ncsu.edu/students/career-services/> are available through the Career Development Center, 2100 Pullen Hall. <https://careers.ncsu.edu/> Make appointments through ePACK <https://ncsu-csm.symplicity.com/students/>

¹ Both electronic and hard-copy, including (but not limited to), copies of lecture notes, exams, answers to exam questions, or any handouts provided by the Instructor.

² Any person not signed up for the course, a personal, public, or commercial website, or any other news or advertising media.